

QUESTION PAPER

Name of the Examination: WINTER 2022-2023-CAT 1

Course Code: CSE2008

Course Title: Operating Systems

Set number: 1

Date of Exam: 15-02-2023 (FN)(C1)

Duration: 90 Minutes

Total Marks: 50

Instruction: Answer ALL the questions.

Q1. Write short notes on following:

- a. Virtualization
- b. Time sharing operating system

[10 Marks]

Q2. Assuming that the system call `fork()` never fails, consider the following C programs P1 and P2 executed on a Unix/Linux system: **[10 Marks]**

<pre>/*P1*/ int main(){ fork(); fork(); fork(); printf("VIT-AP\n"); }</pre>	<pre>/*P2*/ int main(){ fork(); printf("VIT-AP\n"); fork(); printf("VIT-AP\n"); fork(); printf("VIT-AP\n"); }</pre>
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What is the output for the above-mentioned code and justify your answer with the process tree.

Q3. Draw Gantt charts for the following set of processes using the following scheduling algorithms: (a) FCFS (b) SJF (Preemptive and Non-preemptive). Compute the waiting time and turnaround time for each process assuming the context switch time as 0. What are the average values for waiting time and Turnaround time? How many context switching decisions are made in each case (not counting initial decision to give the CPU to P1)?

[10 Marks]

Process	Burst Time	Arrival Time
P1	8	0
P2	5	1
P3	3	2
P4	1	3
P5	7	4

- Q4.** Assume the following processes that arrive for execution at the time indicated and also the length of the CPU burst time is given in ms. **[10 Marks]**

Process	Burst Time	Priority	Arrival Time
P1	10	5	0
P2	6	2	0
P3	7	4	1
P4	4	1	1
P5	5	3	2

- a) Give a Gantt chart illustrating the execution of these processes using Round robin (Quantum=5), and Priority (Preemptive and Non preemptive)
- b) Calculate the average TAT and WT for the above scheduling algorithm.

- Q5.** What are the two models of inter process communication? What are the strengths and weaknesses of the two approaches? **[10 Marks]**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1,2	1	1	10
Q2	1	1	1,2	1	1	10
Q3	2	4	2,5	4	1,2,3,4	10
Q4	2	4	2,5	4	1,2,3,4	10
Q5	2	2	1,2,4	2,3	1,2	10

QUESTION PAPER

Name of the Examination: WINTER 2022-2023-CAT-1

Course Code: CSE-2008

Course Title: Operating System

Set number: 3

Date of Exam: 17-02-2023 (FN)

Duration: 90Mins

Total Marks: 50 (E1)

Instructions:

1. Answer all the Questions

- Q1. a)** Explain in detail the role of Operating system as a resource Manager.
b) Discuss the essential properties of Time sharing and distributed systems.

[5M]

[5M]

- Q2.** Consider the following code segment.

[10M]

```
#include <stdio.h>
#include <sys/types.h>
int i;
int main()
{
    printf("Hello\n");
    fork();

    for(i=0;i<=3;i++)
    {
        printf("%d\n",i);
    }
}
```

What is the output for the abovementioned code and justify your answer with the process tree.

- Q3.** Consider the following table for arrival times and burst times for five processes:

[10M]

Process ID	Arrival Time	Burst Time
P1	5	8
P2	4	5
P3	1	2
P4	0	4
P5	2	6

Calculate the average waiting time and average turnaround time using Shortest Job First and Shortest Remaining Job First scheduling algorithms.

Q4. Consider the following table for arrival times and burst times for five processes;

[10M]

Process ID	Arrival Time	Burst Time	Priority
P1	0	6	2
P2	1	4	1
P3	2	2	3
P4	3	3	1
P5	4	5	4

Calculate the average waiting time and average turnaround time using Priority scheduling and longest job first scheduling algorithms.

Q5. Illustrate how two processes can communication with each other using message queue using suitable example. **[10 M]**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1,2	1	2	1	10
Q2	1	1	1,2	2	1	10
Q3	2	2	1,2	2	1	10
Q4	2	2	4	-	1	10
Q5	2	2	4	-	1	10

QUESTION PAPER

Name of the Examination: WINTER 2022-2023 – CAT-1

Course Code: CSE2008

Course Title: Operating Systems

Set number: 6

Date of Exam: 17-02-2023 (AN)

Duration: 90 Mins

Total Marks: 50 (E2)

All questions carry equal marks.

Q1. a) Consider the following piece of C code

```
#include <stdio.h>
#include <sys/types.h>
int main()
{
    fork();
    fork();
    fork();
    printf("hello\n");
    return 0;
}
```

What is the output of the program, and how many child processes are created during the execution of this program? Justify your answer. (5M)

b) What is a System call? Explain the various types of system calls provided by an operating system. (5M)

2. Direct memory access is used for high-speed I/O devices in order to avoid increasing the CPU's execution load.

- How does the CPU interface with the device to coordinate the transfer?
- How does the CPU know when the memory operations are complete?
- The CPU is allowed to execute other programs while the DMA controller is transferring data. Does this process interfere with the execution of the user programs? If so, describe what forms of interference are caused. (10M)

3. How cooperating processes can communicate in a shared-memory environment? Explain it with a solution to the producer-consumer problem of having a bounded buffer area. (10M)

4. Consider the set of 5 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time	Priority
P1	0	4	2
P2	1	3	3

Process Id	Arrival time	Burst time	Priority
P3	2	1	4
P4	3	5	5
P5	4	2	5

Calculate the average waiting time and average turnaround time for Priority scheduling preemptive mode and non-preemptive mode. (The higher number represents higher priority) (10M)

5. Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process	Burst Time	Arrival Time
P1	5	0
P2	3	1
P3	1	2
P4	2	3
P5	3	4

The processes are assumed to have arrived in the order P1, P2, P3, P4 and P5 all at time 0.

a) Draw Three Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF(Non-Preemptive), RR (time quantum = 2).

b) What is the turnaround time of each process for each of the scheduling algorithms in part a?

c) What is the waiting time of each process for each of these scheduling algorithms?

d) Which of the algorithms results in the minimum average waiting time (over all processes)?

(10M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1,2	1,2	1,2	1,2	10
Q2	1	1	1,2	1	1	10
Q3	2	4	2,5	4	1,2,3,4	10
Q4	2	4	2,5	4	1,2,3,4	10
Q5	2	4	2,5	4	1,2,3,4	10